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four current:-

$J^\mu (\rho c, \vec{j})$

$J_\mu (\rho c, -\vec{j})$ (minkoski space)

Contravariant:-

$\nabla_\mu J^\mu = 0$ or ∂

Electromegnetic four potatial

$A^\mu \left(\frac{\phi}{c}, \vec{A}\right)$

Electromegnatic field Tensor

$F^{\alpha\beta} = \nabla^\alpha A^\beta - \nabla^\beta A^\alpha$

$F_{\alpha\beta} = g_{\alpha\nu} F^{\nu\sigma} g_{\sigma\beta}$

Maxwell field eq. in tensor

$F_{\alpha\beta,\gamma} + F_{\beta\gamma,\alpha} + F_{\gamma\alpha,\beta} = 0$

Electromegnetic energy momentum Tensor

$T^\alpha_\beta = \frac{1}{4} \delta^\alpha_\beta (F_{\mu\nu} F^{\mu\nu}) - F_{\mu\beta} F^{\mu\alpha}$

Momentum Tensor in flat space:-

$T^\alpha_\beta = \frac{1}{4} \delta^\alpha_\beta (F_{\mu\nu} F^{\mu\nu}) - F_{\mu\beta} F^{\mu\nu}$

Mean Stress:-

$\sigma_m = \frac{1}{3} T_r(\sigma_{ij}) = \frac{1}{3} (\sigma_1 + \sigma_2 + \sigma_3)$

mean Pressure:-

$\vec{p} = -\sigma_m = -\frac{1}{3} (\sigma_1 + \sigma_2 + \sigma_3)$

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Reimann Curvature:-

$$K = K(x^i, u^i, v^i) = \frac{R_{ijkl} u^i v^j u^k v^l}{G_{pqrs} u^p v^q u^r v^s}$$

$$G_{pqrs} = g_{pr} g_{qs} - g_{ps} g_{qr}$$

Einstein Tensor:-

$$G^i_j = R^i_j - \frac{1}{2} \delta^i_j R$$

Energy momentum tensor:-

$$\nabla_i T^{ij} = 0$$

Energy momentum tensor of perfect fluid

$$T^{ij} (\rho + \frac{p}{c^2}) u^i u^j - p g^{ij}$$

Einstine field equation:-

$$(R^{\alpha\beta} - \frac{1}{2} g^{\alpha\beta} R)_{;\beta} = 0$$

Maxwell field equation:-

$$\nabla \cdot E = \frac{\rho}{\varepsilon_0} \qquad \nabla \cdot B = 0$$

$$\nabla \times E = -\frac{\partial B}{\partial t} \qquad \nabla \times B = \mu_0 j + \frac{1}{c^2} \frac{\partial E}{\partial t}$$

∇

Charge Density

$$\rho_0 = \frac{Ne}{l_0^3} \qquad \rho = \frac{\rho_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Current Density:-

$$\vec{j} = \rho \vec{v}$$

$$j = \frac{\rho_0 \vec{v}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

mass:-

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Date: Day:

Instantaneous Rest frame:-

$$\vec{a} = \sqrt{a_x^2 + a_y^2 + a^2 z} = \alpha$$

Relativistic Mass:-

$$\vec{m} = \frac{m}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Relativistic momentum:-

$$(p^i) = (p^0, \vec{P}) = (\vec{m} v_i)$$

$$F = \frac{d}{dt}\left(\frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}}\right)$$

minkoski force

$$F = \frac{Fi}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Reimann Tensor:-

$$((v_i)_{;j})_{;k} = v_{i;jk}$$

$$R_{ijkl} = g_{ir} R^r_{jkl}$$

number of Riemann tensor

$$\frac{n^2(n^2-1)}{12}$$

Ricci Tensor

$$R_{ab} = R^c_{acb}$$

Ricci Scaler

$$R = g^{ab} R_{ab} = R^a_a$$

Date: Day:

Simple matrix form 4x4

$$A = \begin{pmatrix} a & b & 0 & 0 \\ b & a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{where } a^2 - b^2 = 1$$

Relative velocity =

$$\beta = -b/a$$

four velocity acceleration:

$$v_i = \frac{dx^i}{dt} = (c, \vec{v})$$

where $\vec{v} = \frac{d\vec{x}}{dt}$ and $\|v\| = \sqrt{v_x^2 + v_y^2 + v_z}$

Lorentz Transformation of Acceleration:-

$$\bar{a}_x = \frac{a_x (1-\beta^2)^{3/2}}{\left(1 - \frac{v a_x}{c^2}\right)^3}$$

Proper Time:-

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}}$$

Proper velocity and acceleration

$$x^i = \frac{dx^i}{d\tau}, \quad b^i = \frac{du^i}{d\tau} = \frac{d^2 x^i}{d\tau^2}$$

Date: Day:

Geodesics:-

$$\frac{d^2x^i}{ds^2} + \Gamma^i_{pq} \frac{dx^p}{ds} \frac{dx^q}{ds} = 0$$

$$x^i = x^i(s)$$

Absolut Derivative:-

$$\frac{\delta T}{\delta t} = \frac{d}{dt}(T^i) + \Gamma^i_{jk} T^j \frac{dx^k}{dt}$$

Reletvistic length:-

$$s^2 = (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2$$

General Length:- (Minkowski spacetime)

$$(ds)^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^3$$

Lorentz Matrices and Transformations:-

Invariance of equation:-

$$g_{ij} d\bar{x}^i d\bar{x}^j = 0 = g_{ij} dx^i dx^j$$

Metric form

$$A^T G A = G$$

Simple Lorentz Transformation:-

$$\bar{x}^0 = \frac{x^0 - \beta x^1}{\sqrt{1-\beta^2}}$$

$$\bar{x}^1 = \frac{-\beta x^0 + x^1}{\sqrt{1-\beta^2}} \qquad \bar{x}^2 = x^2, \quad \bar{x}^3 = x^3$$

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Covariant:-

$$div(A_i) = A_{i;i} = A_{i,i} - \Gamma^{k}_{ii} A_k$$

Curl:-

$$[\nabla \times A]_i = \varepsilon_{i;k} A_{k;j} \quad \text{where}$$

$$A_{k;j} = A_{k,j} - \Gamma^{s}_{kj} A_s$$

Laplacian:-

$$\nabla \cdot \nabla \cdot \nabla \phi = \left(\frac{\partial \phi}{\partial x^i}\right)_{;i}$$

Torsion Tensor:-

$$T^{\alpha}_{\beta\gamma} = \Gamma^{\alpha}_{\beta\gamma} - \Gamma^{\alpha}_{\gamma\beta}$$

Acceleration in general Co-ordinates:-

$$(\vec{a}) = (a^i) = \left(\frac{d}{dt}\left(\frac{dx^i}{dt}\right)\right) = \frac{d^2x^i}{dt^2}$$

instantaneous acceleration of particle.

$$a = \sqrt{\delta_{ij}\, a^i a^j}$$

and $a = \sqrt{g_{ij}\, a^i a^j}$

curvature in general Co-ordinates.

Generalized formula = $\kappa(s) = \sqrt{g_{ij}\, b^i b^j}$

$$s(t) = \int_a^t \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}}\, dt$$

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formulas:-

Khrischoffis Symbol
1st Kind

$$\Gamma_{ijk} = \frac{1}{2}\left[\frac{\partial}{\partial x_i}(g_{jk}) + \frac{\partial}{\partial x^j}(g_{ki}) - \frac{\partial}{\partial x_k}(g_{ij})\right]$$

2nd Kind:-

$$\Gamma^i_{jk} = g^{ir}\Gamma_{jkr}$$

Covariant Differencialion:-

Covariant tensor

$$T_{ijk} = \frac{\partial}{\partial k}(T_i) - \Gamma^t_{ik}(T_t)$$

Contravariant vector Tensor:-

$$T^i_{jk} = \frac{\partial}{\partial x}(T^i) + \Gamma^i_{jk}T^t$$

Gradiant:- of ϕ

$$\Delta\phi = \text{grad}\,\phi = \frac{\partial\phi}{\partial x_i}(\hat{e}_i)$$

Divergence of Contravariant vector

$$\text{div}(At) = A^i_{;i} = A^i_{;i} + \Gamma^i_{ik}AK$$